



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2007
YEAR 11

ASSESSMENT TASK #1

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 67

- Attempt questions ~~1-4~~ 1-5
- Not all questions are of equal value.
- Full marks may not be awarded for careless or badly arranged work.

Examiner: F.Nesbitt

Baros
Cheng
Evans
Ward
McA

QUESTION 1 (12 marks)

- (a) Factorise fully: $4x^2 - 36$ 1
- (b) Evaluate: $\log_5 10$ correct to 2 decimal places. 1
- (c) (i) Express $0.\dot{4}\dot{3}$ as a geometric series, stating the first term and common ratio. 1
- (ii) Use the limiting sum of the series to express $0.\dot{4}\dot{3}$ as a fraction. 2
- (d) Solve $4^x - 9(2^x) + 8 = 0$ 3
- (e) If $f(x) = x^3 - 3x^2 - 24x - 36$, find $f'(-1)$ 2
- (f) Sketch $y = e^x$, marking any intercept(s). 1
- (g) Find $\lim_{x \rightarrow \infty} \frac{2x^2 - 4x}{x^2 - 4}$ 1

QUESTION 2 (14 marks)

- (a) Differentiate (without simplifying):
- (i) $3x\sqrt{x}$ 1
- (ii) $\frac{3x}{x^3 - 1}$ 2
- (iii) $\frac{1}{(3x - 2)^4}$ 2
- (b) If the roots of the quadratic $x^2 - 8x + 5 = 0$ are α and β , find:
- (i) $\alpha + \beta$ and $\alpha\beta$ 1
- (ii) $\alpha^2 + \beta^2$ 2
- (iii) $\alpha^3 + \beta^3$ 2
-
- (c) If $A(x - 1)^2 + Bx + C \equiv x^2$, find A, B and C 2
- (d) (i) Graph $y = 4 - x^2$, showing x and y intercepts 1
- (ii) Hence or otherwise solve $4 - x^2 \geq 0$ 1

QUESTION 3 (13 Marks)

- (a) Given the equation $12y = x^2 - 4x - 32$
- (i) Write the equation in the form: $(x - h)^2 = 4a(y - k)$. 2
 - (ii) Find the coordinates of the focus and vertex. 2
 - (iii) Find the equation of the axis of symmetry. 1
 - (iv) Find the x intercepts. 2
 - (v) Sketch the parabola. 1
- (b) The sum of the first two terms of a geometric series is 15 and the third term is 20.
Find the first term and the common ratio if the series does not have a limiting sum. 3
- (c) If the events A and B are mutually exclusive and $P(A) = 0.3$ and $P(B) = 0.5$
Find:
- (i) $P(A \cup B)$
 - (ii) $P(A \cap B)$ 2

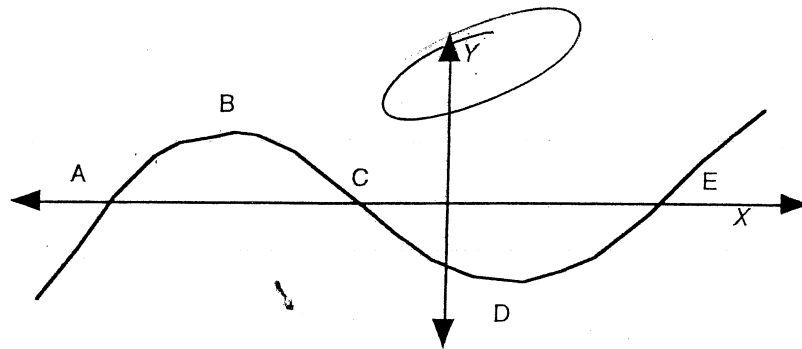
QUESTION 4 (13 Marks)

Solve for x :

- (a) $\log_2(x + 4) - \log_2(x - 2) = 1$ 2
- (b) $4^x = 5^{2x-1}$ correct to 2 decimal places. 3
- (c) $y = f(x)$ is the equation of a continuous function which passes through the point $(0, 3)$.
Find its stationary points and determine their nature if $f'(x) = (x + 1)(x - 3)$ 3
- (d) The tangent and normal to the curve $y = x^3 - x^2 - 6$ at the point $P(2, -2)$ cut the x axis at the points T and N respectively. Find the distance TN. 5

QUESTION 5 (15 Marks)

- (a) Below is a sketch of the gradient function of $f(x)$



- (i) Copy the sketch into your booklet.

What can be said about the behaviour of the curve $y = f(x)$ at the points:

- | | |
|---|---|
| (ii) A, C and E? | 1 |
| (iii) B and D? | 1 |
| (iv) Sketch a possible curve $y = f(x)$, on the same set of axes as (i). | 2 |

- (b) \$60 000 is borrowed at 9% p.a. reducible monthly. Fixed repayments are made at the end of each month for ten years.

- (i) Using M for each monthly repayment, show that the amount owing at the end of the first month,

$$A_1 = 60000 \times 1.075 - M$$

- | | |
|---|---|
| (ii) Write an expression for A_2 , the amount owing at the end of the second month. | 1 |
| (iii) Find an expression for A_n and, using the sum of terms of an arithmetic series or otherwise find the value of M . | 3 |

GBOURTHIC

- (c) Ship A was 60 km due East of Ship B. Ship A sails at 20 km/h due West and ship B sails due South at 30 km/h.

- | | |
|--|---|
| (i) (a) After 30 minutes how far apart are the ships? (nearest km) | 1 |
| (ii) (b) Write an expression for the distance between the ships after x hours. | 2 |
| (iii) Find the value of x when the ships are closest together. | 3 |

Question

$$(a) 4x^2 - 36 = 4(x^2 - 9) \\ = 4(x-3)(x+3)$$

$$(b) \log_5 10 = \frac{\log_{10} 10}{\log_{10} 5} \\ = \frac{1}{0.6989} \\ = 1.43$$

$$(c) (i) 0.\dot{4}3 = 0.43 + 0.0043 + 0.000043 + \dots \\ a = 0.43 \text{ and } r = 0.01$$

$$(ii) S_{\infty} = \frac{a}{1-r} \\ = \frac{0.43}{0.99} \\ = \frac{43}{99}$$

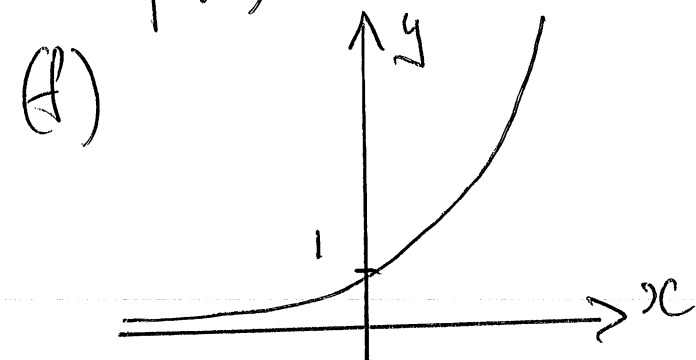
$$(d) 4^x - 9 \cdot 2^x + 8 = 0 \rightarrow \text{let } a = 2^x \text{ then we have:}$$

$$a^2 - 9a + 8 = 0$$

$$(a-8)(a-1) = 0$$

$$\therefore a = 8, 1 \rightarrow 2^x = 8, 1 \\ \text{i.e. } x = 3, 0.$$

$$(e) f(x) = x^3 - 3x^2 - 24x - 36 \\ f'(x) = 3x^2 - 6x - 24 \\ \therefore f'(-1) = 3 + 6 - 24 = -15$$



$$(g) \lim_{x \rightarrow \infty} \frac{2x^2 - 4x}{x^2 - 4} = \lim_{x \rightarrow \infty} \frac{2 - \frac{4x}{x^2}}{1 - \frac{4}{x^2}} \quad \text{and} \quad \left. \begin{matrix} \frac{4x}{x^2} \\ \frac{4}{x^2} \end{matrix} \right\} \rightarrow 0 \text{ as } x \rightarrow \infty \\ = 2$$

Question (2)

(a) (i) $3x\sqrt{x}$
 $= 3x^{3/2}$
 $\frac{d}{dx}(3x\sqrt{x})$
 $= \frac{9}{2}x^{1/2}$ 1

(ii) $\frac{d}{dx}\left(\frac{3x}{x^3-1}\right)$
 $= \frac{3(x^3-1) - 3x(3x^2)}{(x^3-1)^2}$
 $= \frac{3x^3 - 3 - 9x^3}{(x^3-1)^2}$
 $= \frac{-3 - 6x^3}{(x^3-1)^2}$ ✓
 $= \frac{-3(1+2x^3)}{(x^3-1)^2}$

$(3x)(x^3-1)^{-1}$ 2
 $= 3(x^3-1)^{-1} + (3x)(-1)(x^3-1)^{-2} \cdot 3x^2$
 $= \frac{3}{x^3-1} - \frac{9x^3}{(x^3-1)^2}$

(ii) $\frac{d}{dx}(3x-2)^{-4}$
 $= -4(3x-2)^{-5} \times 3$
 $= \frac{-12}{(3x-2)^5}$ 2

(b) $x^2 - 8x + 5 = 0$

1 (i) $\alpha + \beta = 8$ $\alpha\beta = 5$

2 (ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= 64 - 10$
 $= 54$

2 (iii) $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$
 $= 8[54 - 5]$
 $= 8 \times 49$
 $= 392$

14

(c) $A(x-1)^2 + Bx + C \equiv x^2$

put $x = 1$ 2
 $B + C = 1$ (1)

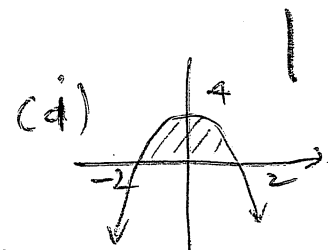
put $x = 0$
 $A + C = 0$
 $C = -A$ (2)

coeff of $x^2 = A = 1$

$\therefore C = -1$

$\therefore B = 2$

(d)



(ii) $-2 \leq x \leq 2$ 1

9283 9614

Question 3

Q1) i $12y = x^2 - 4x - 32$
 $x^2 - 4x + 4 = 12y + 32 + 4$
 $(x-2)^2 = 12y + 36$

$$(x-2)^2 = 12(y+3)$$

$$\boxed{(x-2)^2 = 4 \cdot 3(y+3)}$$

$$h = 2$$

$$a = 3$$

$$k = -3$$

ii focus $(h, k+a)$

$$= (2, -3+3)$$

$$= \boxed{(2, 0)}$$

vertex (h, k)

$$= \boxed{(2, -3)}$$

iii Axis of symmetry

$$\boxed{x = 2}$$

iv x intercepts

$$x^2 - 4x - 32 = 0$$

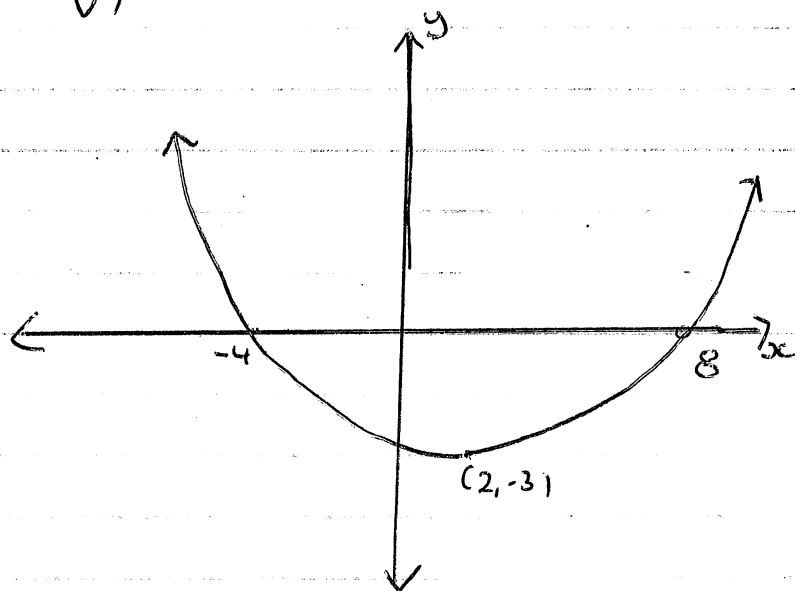
$$(x-8)(x+4) = 0$$

$$\therefore x = 8, -4$$

Pts of intersection

$$\boxed{(8, 0) \text{ and } (-4, 0)}$$

v)



$$b) T_3 = 20$$

$$20 = ar^2 \Rightarrow a = \frac{20}{r^2} \quad (1)$$

$$15 = a + ar \quad (2)$$

sub (1) into (2)

$$15 = \frac{20}{r^2} + \frac{20r}{r^2}$$

$$15r^2 = 20 + 20r$$

$$3r^2 - 4r - \overset{-12}{4} = 0$$

$$\frac{(3r - 6)(3r + 2)}{3} = 0$$

$$(r - 2)(3r + 2) = 0$$

$$\therefore \boxed{r = 2} \quad 3r = -2$$

$$r = -2/3$$

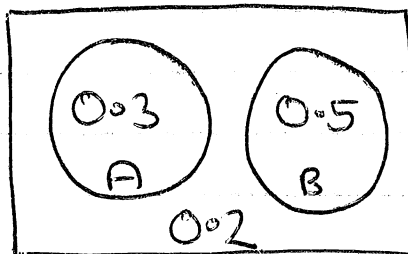
$$\rightarrow |r| > 1 \quad \therefore r \neq -2/3$$

sub $r = 2$ into (1)

$$20 = a \times 2^2$$

$$\boxed{a = 5}$$

c) A & B are mutually exclusive



$$P(A \cup B) = 0.3 + 0.5 = 0.8$$

$$P(A \cap B) = 0$$

SECTION 4.

$$a) \log_2(x+4) - \log_2(x-2) = 1.$$

$$\log_2\left(\frac{x+4}{x-2}\right) = \log_2 2.$$

$$\frac{x+4}{x-2} = 2.$$

$$x+4 = 2x-4.$$

$$\underline{\underline{x=8}}$$

$$b) 4^x = 5^{2x-1}.$$

$$\log_4 4^x = \log_4 5^{2x-1}.$$

$$x \log_4 4 = (2x-1) \log_4 5.$$

$$\frac{x}{2x-1} = \log_4 5.$$

$$x = 2x-1 \left(\frac{\log_{10} 5}{\log_{10} 4} \right)$$

$$+x = 2x-1 (1.16096)$$

$$x = 2x(1.16096) - (1.16096)$$

$$1.3219x = 1.16096$$

$$x = 0.878.$$

$$\underline{\underline{x=0.88 \text{ (2 dp)}}}$$

c) cont)

Points are given by

$$y = \int x^2 - 2x - 3. dx.$$

$$y = \frac{x^3}{3} - x^2 - 3x + C.$$

which passes thro (0,3).

$$y = 0 - 0 - 0 + C.$$

$$\underline{\underline{3=C}}.$$

$$\text{when } x=-1. \quad y = -\frac{1}{3} - 1 + 3 + 3 = 4\frac{2}{3}.$$

A stationary point occurs at $(-1, 4\frac{2}{3})$.

$$\text{when } x=3 \quad y = \frac{27}{3} - 9 + 9 + C.$$

$$y = -6$$

Another stationary point occurs at $(3, -6)$.NATURE

$$\frac{d^2y}{dx^2} = 2x - 2.$$

$$\text{when } x=-1.$$

$$\frac{d^2y}{dx^2} = -4 < 0$$

∴ maximum at $(-1, 4\frac{2}{3})$

$$\text{when } x=3$$

$$\frac{d^2y}{dx^2} = 4 > 0.$$

∴ minimum at $(3, -6)$

$$c) f'(x) = (x+1)(x-3).$$

Stationary points occur when

$$f'(x) = 0$$

$$\therefore f'(x) = (x+1)(x-3) = 0.$$

$$x = -1, +3$$

d) Equation of tangent
at (2, -2) where.

$$y = x^3 - x^2 - 6$$

$$m_T = \frac{dy}{dx} = 3x^2 - 2x \text{ when } x=2.$$

$$\frac{dy}{dx} = 3(4) - 4 = 8.$$

Point gradient formula.

$$y - y_1 = m(x - x_1).$$

$$y - -2 = 8(x - 2).$$

$$y = 8x - 18$$

This crosses x axis at $y=0$

$$0 = 8x - 18.$$

$$8x = 18$$

$$\underline{\underline{x = 2.25.}}$$

Equation of normal.

$$m_N = -\frac{1}{m_T} = -\frac{1}{8}.$$

Point gradient (2, -2) $m = -\frac{1}{8}.$

$$y - -2 = -\frac{1}{8}(x - 2)$$

$$y = -\frac{x}{8} - \frac{1}{4}.$$

when $y=0$. $\frac{x}{8} = -\frac{1}{4}.$

$$\underline{\underline{x = -14}}$$

∴ Distance between.

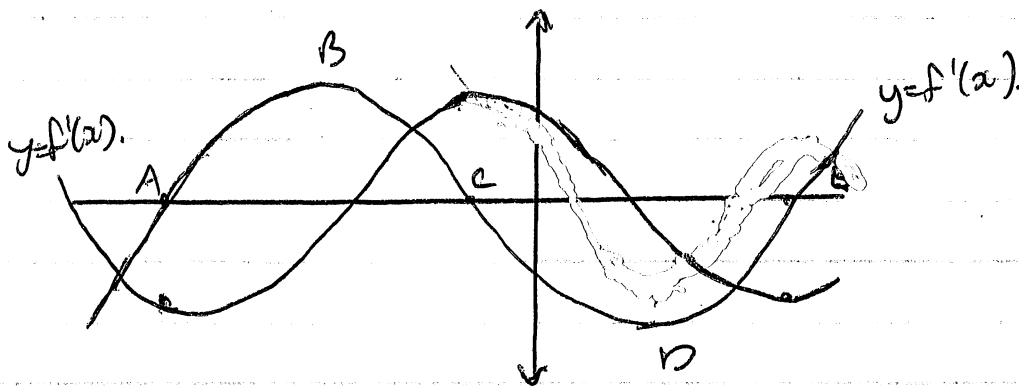
T (2.25, 0) and N (-14, 0)

is 16.25 units

MATH Task 1.

QUESTION 5

a i)



i) $y=f(x)$ has stationary (turning) points at A, C and E.

ii) The curve $y=f(x)$ has points of inflection at B and D.

~~6 i)~~

$$A_1 = 60000 \left(1 + \frac{r}{1200} \right) - M$$

$$= 60000(1.0075) - M.$$

ii)

$$A_2 = A_1(1.0075) - M$$

$$= 60000(1.0075)^2 - (1.0075)M - M.$$

iii)

$$A_n = PR - M$$

$$A_2 = PR^2 - MR - M.$$

$$A_3 = PR^3 - MR^2 - MR - M.$$

$$A_n = PR^n - MR^n - MR^{n-1} - \dots - MR - M.$$

$$= PR^n - n(R^{n-1} + R^{n-2} + \dots + MR + M).$$

$$= PR^n - M \left(\frac{1-R^n}{1-R} \right).$$

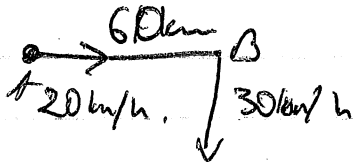
when $A_{120} = 0$.

$$0 = 60,000(1.0075)^{120} - M \left(\frac{1 - 1.0075^{120}}{1 - 1.0075} \right)$$

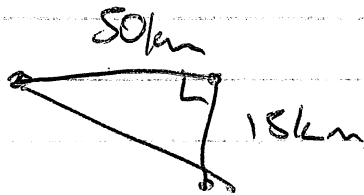
$$M = 60,000(1.0075)^{120} \times \frac{0.0075}{1 - 1.0075^{120}}$$

$$= \$760.05.$$

C(i)



$$AB^2 = 50^2 + 12^2 \\ = 2725$$



$$AB \approx 52 \text{ km.}$$

$$(ii) AB^2 = (60 - 20x)^2 + (30x)^2$$

$$= 3600 - 2400x + 400x^2 + 900x^2$$

$$AB = 10\sqrt{13x^2 - 24x + 36}$$

(ii)

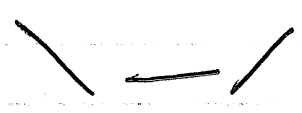
$$\frac{dAB}{dx} = 5(13x^2 - 24x + 36)^{-\frac{1}{2}} (26x - 24)$$
$$= \frac{5(26x - 24)}{\sqrt{13x^2 - 24x + 36}}$$

Set ph $\frac{dAB}{dx} = 0$. ~~0~~ $0 = \frac{x}{y}$ When $x=0$.

$$26x - 24 = 0$$
$$x = \frac{12}{13}$$

x	0	$\frac{12}{13}$	1
$\frac{dAB}{dx}$	-20	0	2

$$\frac{10}{\sqrt{25}} \neq$$



$\therefore x = \frac{12}{13}$ is a minimum

$x = \frac{12}{13}$ hours.